Explainable Deep Learning for Multi-Class Skin Lesion Classification Using EfficientNetV2 and Grad-CAM++ Visualization.

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Beatus Mbunda
Department of Computer Science and Engineering
Jawaharlal Nehru Technological University Hyderabad
Telengana.
India.

bmbunda9@gmail.com

Perumalla Swetha
Department of Computer Science and Engineering
Jawaharlal Nehru Technological University Hyderabad
Telengana
India.

drpswetha@jntuh.ac.in

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***Abstract*—Skin cancer is among the most frequently diagnosed malignancies worldwide, reinforcing the need for early and accurate diagnostic methodologies. Dermoscopic imaging, when integrated with deep learning, has significantly improved diagnostic accuracy. However, challenges persist in model interpretability and optimization. This study proposes an explainable deep learning framework based on EfficientNetV2-M, a lightweight convolutional neural network (CNN) with compound scaling efficiency. The HAM10000 dataset was employed for training with extensive image augmentations, achieving a weighted average accuracy of 96%, precision of 96.17%, recall of 95.84%, and F1-score of 95.82%. Test-Time Augmentation (TTA) using horizontal and vertical flips enhanced model robustness, while a modified classifier head with dropout and batch normalization improved regularization. Explainability was achieved through Grad-CAM++ and Layer-CAM visualizations, highlighting critical dermoscopic regions influencing predictions. The proposed framework demonstrates superior performance and interpretability, promoting reliability and clinical applicability in automated skin lesion diagnosis.**

***Index Terms*—segmentation, Grad-CAM++, Efficient Net v2 M, Test time augmentation, attention mechanism, Skin Cancer.**

I. Introduction

Skin cancer is among the fastest-growing and most frequently diagnosed malignancies worldwide, comprising major types such as malignant melanoma, basal cell carcinoma, and squamous cell carcinoma, each defined by distinct pathological features [1]. Malignant melanoma, originating from melanocytes responsible for melanin production, poses the highest clinical threat due to its aggressiveness and rising global incidence . According to the World Health Organization (WHO), approximately 1,234,533 non-melanoma and 331,722 melanoma cases are reported annually [2]. Dermoscopy has become a fundamental diagnostic tool for dermatological assessment and supports the development of automated detection systems utilizing deep learning techniques , [3]. Datasets such as HAM10000 and ISIC 2019 provide rich, diverse dermoscopic images facilitating model training and benchmarking. Previous studies achieved significant performance using architectures like Inception v3 (96.4% accuracy) [4] CNN–ANN hybrids (92.34%) [5] ,and ResNeXt101 (93.20%) [6] while ResNet101 attained 97.10% accuracy on HAM10000 [7]. Despite such progress, challenges persist including class imbalance, limited labeled data, model overfitting, and lack of interpretability, restricting clinical adoption and generalization. Addressing these issues requires improved data curation, robust validation, and explainable AI (XAI) methods such as Grad-CAM++ visualizations. This study introduces an enhanced EfficientNetV2-M framework for multiclass skin cancer classification on the HAM10000 dataset, integrating attention mechanisms, fine-tuning, and flip-based test-time augmentation to improve diagnostic performance with the Explainable Grad-CAM++ technique. . Experimental outcomes demonstrate superior classification accuracy compared to prior works, validating the model's potential in real-world dermatological applications. The paper is structured into six sections covering introduction, related studies, methodology, experiments, discussion of results, and conclusions.

II. Literature Review

The integration of explainable deep learning in medical image classification, particularly for multi-class skin lesion detection, has gained significant attention in recent years. This section systematically analyzes relevant literature from 2020 to 2025, emphasizing methodologies, performance outcomes, advantages, and limitations. [8] utilized HAM10000 dataset, and DERMNET dataset, with batch size of 64, with epoch 25, and learning rate of 0.001, using MobileNetV2, Resnent50,

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VGG16,VGG19, ResNet50 with best validation accuracy of 94.67%, and training accuracy of 97.57% using the ResNet50 which performs the others model. However the study didn't include explainable AI techniques as well precision, sensitivity and recall to see how the model is performing.

[9] proposed multiclassification model using CNN with GAN based models utilizing DermNet and HAM10000 dataset , with performance of recall 92.7%, F1 score of 92%, accuracy of 93%, and precision of 91.2%. Although the model perform well, it is required high computational power especially GAN model. Furthermore it lacks the explainable mechanism for visualization, which hinder trust for Clinical Experts. [10] utilized HAM10000 dataset with CNN architecture with performance accuracy of 95%. The model sets the foundation for early skin cancer detection and classification. However the model lacks explainable Techniques .

[11] introduced a DenseNet based approach trained on HAM10000, achieving 82.1% accuracy. While effective, DenseNet's memory complexity was a bottleneck in deployment on resource limited devices. In addition the reported performance of the study is limited only to 82.1% accuracy.

[12] utilized dataset from kaggle with 8 classes for skin cancer diseases using Mobilenet V2 and ResNet50 with softmax classifier with corresponding performance of 97.44%, and 95.29% respectively. The study used 925 samples for training and 234 for testing. However the models lack integration of explainable AI mechanism.

III. Methodology

A. Proposed Work and System Architecture

This study proposes an efficient CNN-based framework for automated skin cancer detection using the EfficientNetV2-M model. The approach improves diagnostic accuracy, robustness, and interpretability over the existing benchmark. Experiments were conducted on the HAM10000 dataset containing 10,015 dermoscopic images across seven lesion types . Preprocessing included DullRazor hair removal, data cleaning, normalization, and augmentation (random flips, rotation, Gaussian blur) to match model input requirements.EfficientNetV2-M, pre-trained on ImageNet21K, was fine-tuned on HAM10000 with added dropout and batch normalization to reduce overfitting. Balanced class weighting and Test-Time Augmentation (TTA) improved generalization, while attention mechanisms refined feature extraction.To ensure transparency, Grad-CAM++ and Layer-CAM were used to visualize class-specific regions, providing clinical interpretability . The proposed model achieved 96% accuracy across seven lesion classes, outperforming standard CNNs. Fig. 1 shows the overall system pipeline.

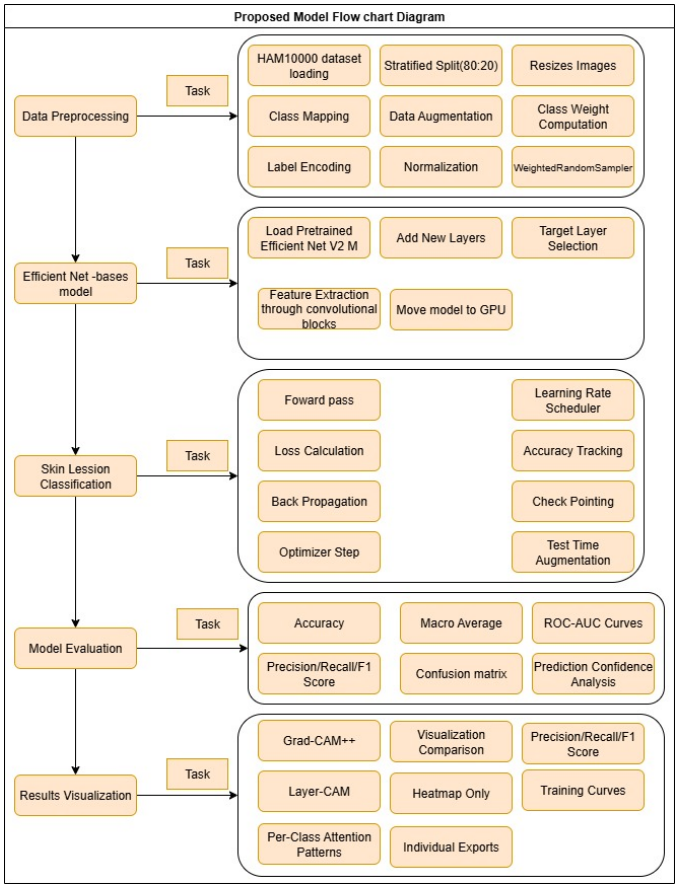


Fig. 1. Architecture Pipeline Flow

B. Data Processing

Image preprocessing: The images were resized to 384x384 which is the default format for Efficient Net V2 M and the model was trained based on pretrained efficient Net V2 . Data augmentation (training only): Through random horizontal/vertical flips help to mimics different orientations of lesions in clinical settings, Preserves malignancy characteristics regardless of orientation, prevents over fitting by training the model that lesion orientation doesn't determine class Rotation (±20 degrees: Help to Handle slight imaging angle variations, Safety .Furthermore limited to 20° to avoid distorting critical diagnostic features like Asymmetry which are important for melanoma detection, and Border irregularities Color jitter help to accounts for color variations in imaging, but also in, medical relevance provides different skin tones which affect lesion appearance, lighting conditions vary across clinics, and Camera sensors have different color responses Random affine transforms help to simulates slight perspective changes, preserves lesion structure while adding viewpoint variation. Random resized crops help to varies lesion to image ratio, teaches model to focus on lesion rather than background. Gaussian blur help to simulates imperfect focus, Forces model to learn robust features beyond sharp edges, mild blurring (kernel=3) preserves diagnostic structures.

C. Algorithms

EfficientNetV2-M is part of the EfficientNetV2 family, an advanced Convolutional Neural Network (CNN) architecture introduced in 2021 to address the limitations of earlier Effi-

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cientNet models. It incorporates progressive learning strategies and an enhanced compound scaling method to simultaneously balance network depth, width, and input resolution more effectively, leading to faster training and improved accuracy In this study, EfficientNetV2-M is employed due to its superior efficiency and performance compared to earlier variants such as EfficientNet-B0 to B4, while maintaining a manageable computational footprint. The experimental setup in this scenario retains the same hyperparameters and loss function used in previous configurations to ensure consistency. However, the input image size is adjusted to meet the optimal resolution requirements of the EfficientNetV2-M model. Table 1 presents hyperparameters, and Fig.2 Proposed Block Diagram

TABLE I
HYPERPARAMETERS USED IN THE PROPOSED MODEL

Hyperparameter	Value
Number of Images	10,015
Image Size	384×384
Splitting Ratio	80:20
Number of Epochs	50
Warm-Up Epochs	10
Training Batch Size	8
Optimizer	AdamW
Visualization	Grad-CAM++, Layer-CAM, Heat Map
Learning Rate Scheduler	Cosine Decay
Testing Strategy	Test-Time Augmentation (TTA)

D. Mathematical Model of the Proposed Framework

The proposed framework combines the EfficientNetV2-M backbone for feature extraction, a modified classification head, and Grad-CAM++ for explainability. The mathematical formulation is presented as

1) Feature Extraction: Given a dermoscopic image

$$x \in \mathbb{R}^{384\times384\times3},$$

the EfficientNetV2-M backbone $f_\theta(\cdot)$ extracts hierarchical feature representations:

$$F = f_\theta(x), \quad F \in \mathbb{R}^{h\times w\times d}$$

where h, w, d denote the spatial height, width, and depth of the feature map.

2) Classification Head: The extracted features are aggregated and passed through fully connected layers with dropout and batch normalization:

$$z = W \cdot \text{GlobalAvgPool}(F) + b$$

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^{K}\exp(z_j)}, \quad c = 1, 2, \ldots, K$$

where $K = 7$ is the number of lesion classes, and p_c is the probability of class c.

3) Loss Function: To handle class imbalance, weighted categorical cross-entropy is employed:

$$\mathcal{L} = -\sum_{c=1}^{K} w_c\, y_c \log(p_c)$$

where y_c is the ground-truth one-hot label, and w_c is the weight for class c.

4) Explainability with Grad-CAM++: The importance weight of activation A_{ij}^k in channel k for class c is:

$$\alpha_{kc}^{(i,j)} = \frac{\frac{\partial^2 y_c}{\partial (A_{ij}^k)^2}}{2\cdot\frac{\partial^2 y_c}{\partial (A_{ij}^k)^2} + \sum_{a,b} A_{ab}^k \cdot \frac{\partial^3 y_c}{\partial (A_{ij}^k)^3}}$$

The Grad-CAM++ localization map is obtained as:

$$L_c^{\text{Grad-CAM++}} = \text{ReLU}\left(\sum_k \sum_{i,j} \alpha_{kc}^{(i,j)} \cdot \frac{\partial y_c}{\partial A_{ij}^k} \cdot A_{ij}^k\right)$$

This highlights the most discriminative regions influencing the model's prediction.

5) Decision Rule: The predicted class is chosen as:

$$\hat{y} = \arg\max_c \; p_c$$

6) Algorithmic Workflow: The overall procedure is summarized in Algorithm III-D6.

[H] EfficientNetV2-M with Grad-CAM++ Dermoscopic image x Predicted class $\hat{y}$, Heatmap L_c Preprocess x (resize, normalize, augment) Extract features $F = f_\theta(x)$ Compute logits z and probabilities p using softmax Compute weighted loss $\mathcal{L}$ Update parameters using backpropagation Compute Grad-CAM++ heatmap L_c for explainability Predict final class $\hat{y} = \arg\max_c p_c$ Table II shows the layer structure

TABLE II
MODIFIED LAYER STRUCTURE FOR THE PROPOSED EFFICIENTNETV2-M CLASSIFICATION HEAD.

Layer name	Layer type	Size of feature map	Activation
Global Avg Pooling 2D	Global Average Pooling 2D	Varies with input resolution	N/A
Dense_1	Dense	512	swish
BatchNormalization_1	BatchNormalization	512	N/A
Dropout_1	Dropout (rate = 0.5)	512	N/A
Dense_2	Dense	256	swish
BatchNormalization_2	BatchNormalization	256	N/A
Dropout_2	Dropout (rate = 0.5)	256	N/A
Dense_output	Dense	7	softmax

for the proposed model classification head.

IV. EXPERIMENTAL RESULTS

Figure 3 shows the prediction summary for class wise. Accuracy refers to a test's ability to correctly identify both positive and negative cases [13]. It is measured by the proportion of true positive and true negative results out of all the cases that were thoroughly analyzed. This might be communicated numerically as presented in equation 1:

$$\text{Accuracy} = \frac{\sum_{k=1}^{K} c_{k,k}}{\sum_{k=1}^{K}\sum_{j=1}^{K} c_{k,j}} \tag{1}$$

Where: $c_{k,k}$ = number of correctly classified samples for class k (diagonal of the confusion matrix); $\sum_{k=1}^{K}\sum_{j=1}^{K} c_{k,j}$ = total number of samples; K = total number of classes.

However focusing on accuracy alone doesn't tell how the model is performing due to imbalance nature of the HAM10000 dataset where majority classes can have high impact for the accuracy than minority classes [14] Furthermore it lacks of clinical interpretability where doctors need to

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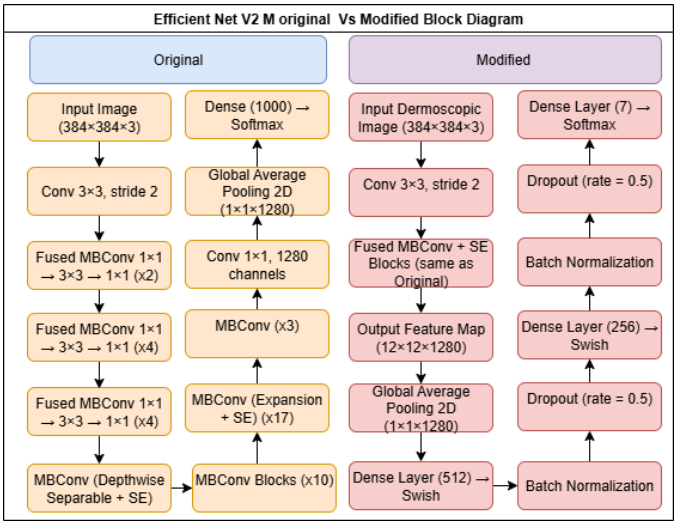


Fig. 2. EfficientNet V2 M Original VS Modified Block Diagram

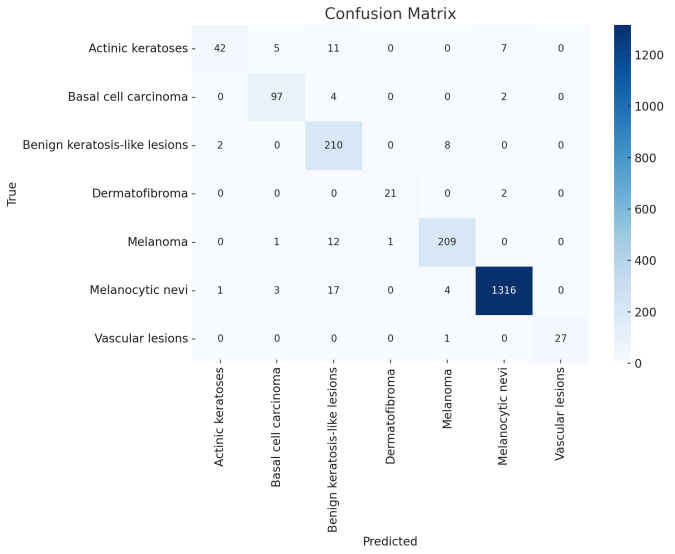


Fig. 3. Confusion Matrix

understand how the decision of the model is being made [15] and [16]. Precision: measures the proportion of correctly identified positive cases out of all those labeled as positive [17]. It calculated using the following formula as written in equation 2:

$$\text{Precision} = \frac{\sum_{k=1}^{K} c_{k,k}}{\sum_{k=1}^{K} \sum_{j=1}^{K} c_{j,k}} \tag{2}$$

Recall: This metric assesses the extent to which a model successfully detects all true instances of a given class [18]. It can be presented using the formula as written in equation 3

$$\text{Recall} = \frac{\sum_{k=1}^{K} c_{k,k}}{\sum_{k=1}^{K} \sum_{j=1}^{K} c_{k,j}} \tag{3}$$

F1-Score: The machine learning metric that combines model precision and recall to assess its performance. It measures how often the model positive prediction are correct across the entire dataset.It calculated as shown in equation 4:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Where:

- $c_{k,k}$: Number of correctly classified samples for class k (diagonal elements of the confusion matrix).
- $\sum_{j=1}^{K} c_{j,k}$: Total number of samples predicted as class k (sum of column k in the confusion matrix).
- $\sum_{j=1}^{K} c_{k,j}$: Total number of actual samples belonging to class k (sum of row k in the confusion matrix).
- K : Total number of classes.

$$\alpha_{kc}^{(i,j)} = \frac{\partial^2 y^c / \partial (A_{i,j}^k)^2}{2 \cdot \partial^2 y^c / \partial (A_{i,j}^k)^2 + \sum_{a,b} A_{a,b}^k \cdot \partial^3 y^c / \partial (A_{i,j}^k)^3} \tag{5}$$

$$L_{\text{Grad-CAM++}}^c = \text{ReLU}\left(\sum_{k,i,j} \alpha_{kc}^{(i,j)} \cdot \frac{\partial y^c}{\partial A_{i,j}^k} \cdot A_{i,j}^k \right) \tag{6}$$

Equation 6 shows the Grad-CAM++ localization formula.

$$L_{\text{Layer-CAM}}^c = \text{ReLU}\left(\sum_{k,i,j} \frac{\partial y^c}{\partial A_{i,j}^k} \cdot A_{i,j}^k \right) \tag{7}$$

Equation 7 shows the Layer-CAM localization formula.

$$\text{Heatmap}(i,j) = \frac{L(i,j) - \min(L)}{\max(L) - \min(L)} \tag{8}$$

Equation 8 shows the heatmap normalization.
Where: y^c = output score for class c before softmax; $A_{i,j}^k$ = activation map value at (i,j) in channel k; $\alpha_{kc}^{(i,j)}$ = weight for (i,j) in channel k for class c (Grad-CAM++); $\partial y^c / \partial A_{i,j}^k$ = first-order gradient; $\partial^2 y^c / \partial (A_{i,j}^k)^2$ = second-order gradient; $\partial^3 y^c / \partial (A_{i,j}^k)^3$ = third-order gradient; $L_{\text{Grad-CAM++}}^c$ = Grad-CAM++ localization map; $L_{\text{Layer-CAM}}^c$ = Layer-CAM localization map; $\text{Heatmap}(i,j)$ = normalized visualization value at pixel (i,j).

V. Results and Discussion

Fig. 3 presents the confusion matrix results for the proposed seven class skin lesion classification system, which was evaluated on a dataset of 2,003 dermoscopic images. The dataset includes the following class distributions: actinic keratoses (akiec) 65 samples, basal cell carcinoma (bcc) 103 samples, benign keratosis-like lesions (bkl) 220 samples, dermatofibroma (df) 23 samples, melanoma (mel) 223 samples, melanocytic nevi (nv) 1,341 samples, and vascular lesions (vasc) 28 samples. The classifier demonstrates excellent performance for several lesion types. Melanocytic nevi (nv), the most prevalent class in the dataset, achieved the highest accuracy, correctly classifying 1,316 out of 1,341 instances

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(98.1%). This reflects robust model generalization for this benign lesion class, which is well represented in the training data. Basal cell carcinoma (bcc) also shows strong performance, with 94.2% accuracy (97/103), indicating reliable detection of this malignant but typically well defined lesion type. Similarly, benign keratosis-like lesions (bkl) were accurately identified in 210 out of 220 cases, corresponding to 95.5% accuracy. Other classes also demonstrate solid classification metrics. Vascular lesions (vasc) achieved 96.4% accuracy (27/28), and dermatofibroma (df) reached 91.3% accuracy (21/23). However, due to the limited sample size for both categories, statistical reliability may be constrained. Nevertheless, these results highlight the model's capacity to generalize effectively even for minority classes. On the other hand, actinic keratoses (akiec) present a significant classification challenge, with the lowest accuracy at 64.6% (42/65 correctly classified). Misclassifications primarily occurred with benign keratosis-like lesions (11 cases) and melanocytic nevi (7 cases), likely due to overlapping morphological characteristics and keratinization patterns. Although melanoma (mel) was classified with a high accuracy of 93.7% (209/223), the presence of 12 false negatives misclassified as bkl raises clinical concerns, particularly given the malignant potential of melanoma. From a clinical perspective, these confusion patterns are noteworthy. The frequent misclassification of akiec as benign lesions may delay intervention for this pre-malignant condition, while incorrect labeling of melanoma poses more serious diagnostic implications due to its aggressive nature. These findings emphasize the need for enhanced sensitivity in detecting early-stage malignancies and pre-malignant lesions. As shown in Figure 3, the classifier achieves outstanding precision, recall, and F1-score for the melanocytic nevi (nv) class, with values of 99.2%, 98.1%, and 98.6%, respectively. These metrics align with the class's dominance in the dataset and confirm earlier findings that models trained on HAM10000 perform exceptionally well on majority classes. Remarkably, vascular lesions (vasc) despite accounting for only 1.4% of the dataset achieved perfect precision (100%) and high recall (96.4%), suggesting that the model has learned effective features for this underrepresented class. In addition, both melanoma (mel) and dermatofibroma (df) exhibit strong F1-scores of 93.7% and 93.3%, respectively, indicating a well-balanced trade off between precision and recall. The benign keratosis like lesions (bkl) class also performs well overall, with an F1-score of 88.6%,driven by a high recall of 95.5%, although its lower precision (82.7%) suggests a tendency for false positives. The basal cell carcinoma (bcc) class maintains a solid F1-score of 92.8%, consistent with its high classification accuracy. In contrast, actinic keratoses (akiec) remains the most problematic class. It achieves the lowest recall (64.6%) and an F1-score of 76.4%, underscoring its susceptibility to misclassification. These results are consistent with earlier observations that akiec shares visual features with mel and bkl, complicating accurate identification. Despite these challenges, the model demonstrates high reliability across both majority and minority classes, highlighting its strong generalization capabilities and potential for clinical deployment.

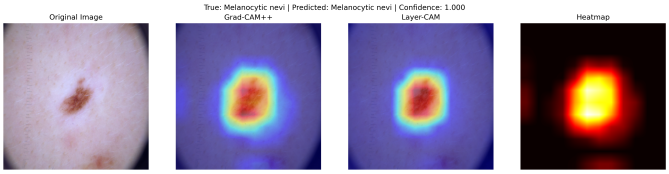


Fig. 4. Melanocytic Nevi Visualization (Original vs Grad-CAM++ vs Layer-CAM vs Heatmap)

A. Analysis

In Fig.4, the original image shows a well-defined, symmetric brown lesion with uniform pigmentation of benign Melanocytic nevus. The Grad-CAM++ visualization highlights a concentrated activation region centered on the lesion, indicating that the model correctly focuses on the lesion core. Layer-CAM exhibits a slightly sharper and more localized activation, with enhanced edge awareness compared to Grad-CAM++, confirming consistent attention across model layers. The heatmap shows a single dominant yellow red activation area corresponding to the lesion region, suggesting that the model's decision is spatially coherent and lesion-focused. .

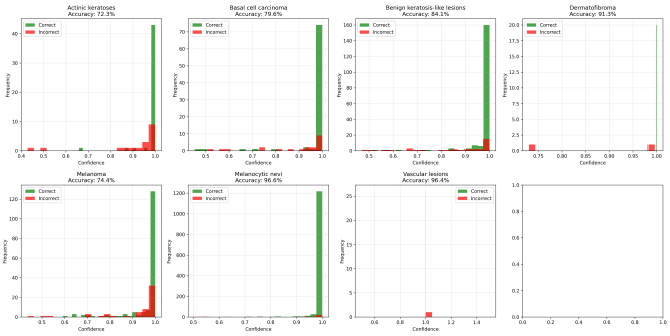


Fig. 5. Confidence Score

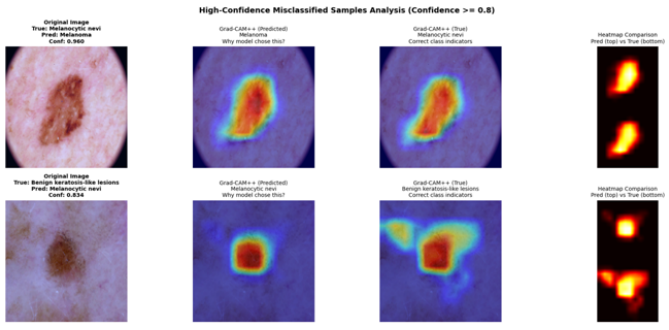


Fig. 6. Misclassified Confidence Score

Fig. 5 presents the distribution of prediction confidences where green color present correct prediction and red color present incorrect prediction classifications across lesion classes, offering insight into model calibration and uncertainty. In addition

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to confidence distribution analysis (Fig. 5), Grad-CAM++ visualizations were generated for high-confidence misclassified samples (Fig. 6) to qualitatively assess the model's failure modes. The confusion matrix presented in Fig. 3 highlights that most misclassifications occurred between visually similar lesion classes, particularly melanocytic nevi and melanoma, as well as benign keratosis-like lesions and actinic keratoses. To further interpret these misclassification patterns, Grad-CAM++ visualizations were applied to high-confidence misclassified samples (confidence 0.8), as shown in Fig. 6. This qualitative assessment reveals the regions most influential to the model's decisions, providing deeper understanding of its attention mechanisms and explaining how visual feature overlap between lesion types contributes to overconfident misclassifications.

For instance actinic keratoses (akiec), with an accuracy of 72.3%, a considerable number of incorrect predictions occur at high confidence levels which is greater than 0.9, indicating overconfidence and poor separability. This due to miscalibration potentially arising from visual similarity with other classes, such as melanoma and benign keratosis-like lesions (bkl). In contrast, basal cell carcinoma (bcc), which achieved 79.6% accuracy, shows most correct predictions concentrated near maximum confidence. However, a moderate number of incorrect predictions also appear at high confidence, indicating tendency for overconfident misclassifications. Benign keratosis-like lesions (bkl), with an accuracy of 84.1%, display a confidence distribution skewed toward high confidence correct predictions. Nevertheless, the presence of high confidence incorrect predictions suggests potential class overlap or labeling inconsistencies. Despite limited support (n = 23), dermatofibroma (df) class exhibits strong performance with 91.3% accuracy. Correct predictions are tightly clustered at confidence levels above 0.95, whereas incorrect predictions are rare and generally assigned lower confidence, indicating reliable calibration for this minority class. Melanoma (mel) presents an accuracy of 74.4% but exhibits numerous incorrect predictions at high confidence. This highlights substantial miscalibration and frequent confusion with visually similar classes such as nevus (nv) and bkl. Calibration refinement and class specific augmentation strategies are recommended.

B. Instance-Based Interpretation of Misclassification

As shown in Fig. 3, the confusion matrix reveals a significant overlap between *melanocytic nevi (nv)* and *melanoma (mel)*, indicating the model's difficulty in distinguishing these visually similar lesion types. This trend is further supported by the class-wise confidence score distributions presented in Fig. 5, where several melanoma samples were misclassified as melanocytic nevi with confidence scores exceeding 0.9.

For instance, one melanoma sample misclassified as *nv* exhibited a prediction confidence of 0.91, reflecting overconfidence despite the incorrect decision. The corresponding Grad-CAM++ visualization in Fig. 6 reveals that the network predominantly focused on uniformly pigmented regions while neglecting irregular structures and asymmetrical features that are characteristic of melanoma. This attention bias suggests that the model relied more on texture homogeneity rather than malignant asymmetries, leading to clinically significant misinterpretations.

Similarly, *actinic keratoses (akiec)* demonstrated frequent high-confidence errors (0.9), where predictions were incorrectly assigned to *benign keratosis-like lesions (bkl)* due to overlapping keratinized surface patterns. This overconfident misclassification emphasizes poor feature separability and miscalibration. In contrast, minority classes such as *dermatofibroma (df)* and *vascular lesions (vasc)* displayed reliable calibration, with incorrect predictions typically occurring at moderate or low confidence levels (¡0.7).

Despite the inherent class imbalance melanocytic nevi accounts for approximately 60% of the dataset the classifier demonstrates robust performance across both majority and minority classes. Notably, dermatofibroma and vascular lesions, which are underrepresented, still achieve excellent AUC scores. This indicates the model's ability to generalize effectively across diverse class distributions.

C. Comparative Analysis with Existing Models

To comprehensively evaluate the effectiveness of the proposed EfficientNetV2-M architecture, a comparative analysis was conducted against several state-of-the-art deep learning and traditional machine learning approaches reported in recent studies. Prior works utilizing EfficientNetV2 variants such as EfficientNetV2-B2 and EfficientNetV2-Small achieved accuracies of 89% and 88.62%, respectively [19], [20]. Other CNN-based architectures including EfficientNetB0, VGG19, MobileNet, and ResNet50V2 reported performance levels around 84% [21], while hybrid frameworks using RegNetY-32, InceptionV3, and AlexNet attained up to 91% accuracy [22]. Conventional CNN models applied to the HAM10000 dataset reached 92% [23], whereas combinations of CNN, ResNet50, and InceptionV3 produced 88.6% accuracy [24]. Traditional machine learning ensembles such as KNN, Decision Tree, and Random Forest obtained 95.5% accuracy [25], and deep transfer-learning architectures involving VGG16, VGG19, and MobileNet variants achieved approximately 92.31% accuracy [26]. In contrast, the proposed modified EfficientNetV2-M model attained 96% accuracy, surpassing prior results by a margin of 4–8% due to incoparating class weighted sampling , test time augmentation, dropout and cosine decay learning rate scheduling. These comparative findings validate the robustness, interpretability, and clinical relevance of the proposed model over existing CNN-based approaches for multi-class skin lesion classification.

VI. CONCLUSION

This study presented a modified EfficientNetV2-M architecture for multiclass skin lesion classification using the HAM10000 dermoscopic dataset. The model was trained and evaluated on 10,015 high quality, preprocessed images, utilizing transfer learning with ImageNet weights to enhance

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generalization and reduce overfitting. The proposed framework demonstrated superior classification accuracy and AUC compared to several state-of the-art CNN architectures. Explainability was integrated through Grad-CAM, Layer-CAM, and heatmap visualizations, providing clear insights into the discriminative regions influencing model predictions. Despite its strong overall performance, the model faced challenges due to class imbalance, particularly the underrepresentation of malignant lesions. Although melanoma achieved a high precision of 93. 7%, certain malignant samples were misclassified as benign keratosis-like lesions, raising concerns about clinical reliability. Similarly, actinic keratoses showed a lower accuracy of 64.6%, primarily due to morphological overlaps with other lesion types. Minority classes such as vascular and dermatofibroma achieved accuracies of 96.4% and 91.3%, respectively, though limited sample sizes restrict statistical robustness. Future work will focus on addressing class imbalance using hybrid resampling and generative methods, incorporating multimodal data integration, and validating the model's diagnostic utility in real-world clinical environments.

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